AN ORDINAL-PRIME CMZ DEFORMATION CRITERION FOR THE RIEMANN HYPOTHESIS

Abstract. We give a self-contained arithmetic-to-dynamical bridge between the Riemann hypothesis and a two-channel logarithmic stretching observable motivated by the multiscale mechanisms of Córdoba, Martínez-Zoroa, and Zheng. The arithmetic input is the sign-specific ordinal-prime transform S, its summatory function SS, and the Dirichlet series α. An exact finite Abel identity, including its unfinished-pair boundary term, identifies the residual Dirichlet polynomial $\delta_Q(\sigma)$ with the logarithm of the relative deformation of two positive CMZ-type channels. We prove that the Riemann hypothesis is equivalent to tail stability of this relative deformation for every real $\sigma > 3/2$. We also record the domains of absolute convergence, all justified interchanges, the meromorphic blocks of α, and the singularities that cannot be transferred to the full series without additional continuation. The equivalence is rigorous; a realisation by the published Euler gluing scheme is stated separately as an open PDE problem and is not asserted here.

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Remark 0.1 (Mathematical status and provenance). Statements marked [LIT] are imported from the cited literature. Statements marked [DED] are proved in this article from the stated definitions. Statements marked [ADAPT] adapt a published mechanism to a different object. A statement marked [OPEN] is not proved. The exact ordinal identities and the CMZ–RH equivalence are [DED]. The scalar stretching law is an [ADAPT] of the CMZ amplitude mechanism. No theorem in this article claims that the existing Euler, Boussinesq, or IPM constructions supply the missing arithmetic tail estimate.

1. Introduction

For $s = \sigma + it$ with $\sigma > 1$, the Riemann zeta function is defined by the absolutely convergent Dirichlet series and Euler product

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_{p} \left(1 - p^{-s}\right)^{-1}. \tag{1.1}$$

It extends meromorphically to $\mathbb{C}$, has a simple pole at $s = 1$, and satisfies a functional equation relating s and $1 - s$; see [3, 1]. Its nontrivial zeros lie in $0 < \Re s < 1$. The Riemann hypothesis is the assertion

$$\mathrm{RH}: \qquad \zeta(\rho) = 0, \quad 0 < \Re\rho < 1 \quad \Longrightarrow \quad \Re\rho = \frac{1}{2}. \tag{1.2}$$

The bridge developed here begins with four classical arithmetic functions. The Möbius function μ, the Liouville function λ, and the von Mangoldt function Λ are

$$\mu(n) = \begin{cases} 1, & n = 1, \\ (-1)^r, & n \text{ is a product of } r \text{ distinct primes}, \\ 0, & p^2 \mid n \text{ for some prime } p, \end{cases} \tag{1.3}$$

$$\lambda(n) = (-1)^{\Omega(n)}, \tag{1.4}$$

$$\Lambda(n) = \begin{cases} \log p, & n = p^k \text{ with } p \text{ prime and } k \geq 1, \\ 0, & \text{otherwise.} \end{cases} \tag{1.5}$$

Here $\Omega(n)$ counts prime factors with multiplicity. Absolute convergence and multiplication of Euler factors give, for $\sigma > 1$,

$$\frac{1}{\zeta(s)} = \sum_{n=1}^{\infty} \frac{\mu(n)}{n^s}, \tag{1.6}$$

$$\frac{\zeta(2s)}{\zeta(s)} = \sum_{n=1}^{\infty} \frac{\lambda(n)}{n^s}, \tag{1.7}$$

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s}. \tag{1.8}$$

The first identity makes cancellation in μ directly sensitive to zeros of ζ. The second and third identities explain why Liouville signs and prime-power weights are natural comparison objects. The construction below uses μ itself; λ and Λ are introductory reference points, not substituted coefficients.

Write

$$M(x) = \sum_{n \leq x} \mu(n). \tag{1.9}$$

The classical Littlewood criterion states

$$\mathrm{RH} \quad \Longleftrightarrow \quad M(x) = O_\eta\left(x^{1/2+\eta}\right) \quad \text{for every } \eta > 0. \tag{1.10}$$

We use this imported theorem in its stated pointwise form [2, 3]. The objective of this article is not to replace (1.10) by an unproved estimate under a new name. It is to identify exactly how an ordinal-prime observable enters a CMZ logarithmic stretching law and to state the resulting equivalence without discarding cut-off or analytic-domain information.

2. The ordinal-prime transform

2.1. **Counters and transformed values.** Let p_j be the jth prime and put $p_0 = 0$. For real $x \geq 1$, define

$$N_+(x) = \#\{n \leq x : \mu(n) = 1\}, \tag{2.1}$$

$$N_-(x) = \#\{n \leq x : \mu(n) = -1\}, \tag{2.2}$$

$$Q(x) = N_+(x) + N_-(x), \tag{2.3}$$

$$M(x) = N_+(x) - N_-(x). \tag{2.4}$$

The last definition agrees with (1.9). In particular,

$$N_\pm(x) = \frac{Q(x) \pm M(x)}{2}. \tag{2.5}$$

Definition 2.1 (Ordinal-prime transform)**.** The sign-specific ordinal-prime transform and its summatory function are

$$S(n) = \begin{cases} p_{N_+(n)}, & \mu(n) = 1, \\ -p_{N_-(n)}, & \mu(n) = -1, \\ 1, & \mu(n) = 0, \end{cases} \tag{2.6}$$

$$SS(x) = \sum_{n \leq x} S(n). \tag{2.7}$$

Thus the first positive Möbius value receives p_1, the second receives p_2, and so on; the negative occurrences receive the negatives of the same prime list on their own clock. The value 1 at nonsquarefree integers is part of the definition and must be retained.

Put

$$\mathcal{P}(m) = \sum_{j=1}^{m} p_j, \qquad \mathcal{P}(0) = 0. \tag{2.8}$$

Proposition 2.2 (Closed form for SS)**.** *For every real $x \geq 1$,*

$$SS(x) = \lfloor x \rfloor - Q(x) + \mathcal{P}\big(N_+(x)\big) - \mathcal{P}\big(N_-(x)\big). \tag{2.9}$$

Consequently, with

$$T(x) = SS(x) - \lfloor x \rfloor + Q(x), \tag{2.10}$$

one has

$$T(x) = \mathcal{P}\big(N_+(x)\big) - \mathcal{P}\big(N_-(x)\big). \tag{2.11}$$

Proof. There are $\lfloor x \rfloor - Q(x)$ nonsquarefree integers up to x, each contributing 1. The positive squarefree occurrences contribute the prime prefix $p_1 + \cdots + p_{N_+(x)}$, and the negative occurrences contribute the negative prime prefix $-(p_1 + \cdots + p_{N_-(x)})$. Their sum is (2.9); subtracting the zero row gives (2.11). □

2.2. **Ordinal and prime-displacement parts.** For a squarefree n, let $j(n)$ denote its occurrence number inside its own sign class. Define

$$w_j = p_j - j, \qquad D_w(m) = \sum_{j=1}^{m} w_j, \qquad D_w(0) = 0. \tag{2.12}$$

Lemma 2.3 (Exact ordinal coefficient sum)**.** *For every real $x \geq 1$,*

$$\sum_{\substack{n \leq x \\ \mu(n) \neq 0}} \mu(n) j(n) = \frac{M(x)(Q(x)+1)}{2}. \tag{2.13}$$

Proof. The positive occurrences contribute $1 + \cdots + N_+$ and the negative occurrences contribute $-(1 + \cdots + N_-)$. Therefore

$$\sum_{\substack{n \le x \\ \mu(n) \neq 0}} \mu(n) j(n) = \frac{N_+(N_+ + 1) - N_-(N_- + 1)}{2} \tag{2.14}$$

$$= \frac{(N_+ - N_-)(N_+ + N_- + 1)}{2} \tag{2.15}$$

$$= \frac{M(x)(Q(x) + 1)}{2}. \tag{2.16}$$

□

Definition 2.4 (Prime-displacement partial sum)**.** Set

$$W(x) = \sum_{\substack{n \le x \\ \mu(n) \neq 0}} \mu(n) w_{j(n)}. \tag{2.17}$$

Proposition 2.5 (Exact recovery of W from SS)**.** *For every real $x \ge 1$,*

$$W(x) = D_w(N_+(x)) - D_w(N_-(x)) \tag{2.18}$$

$$= SS(x) - \lfloor x \rfloor + Q(x) - \frac{M(x)(Q(x) + 1)}{2}. \tag{2.19}$$

Proof. The first equality follows by summing w_j on the two occurrence clocks. Since $p_j = j + w_j$, the signed-prime contribution $T(x)$ is the sum of the ordinal contribution in Lemma 2.3 and $W(x)$. Use (2.10) and rearrange. □

Proposition 2.6 (Asymptotic amplification)**.** *Let $c_0 = 3/\pi^2$. As $x \to \infty$,*

$$W(x) = c_0 x \log x \, M(x)(1 + o(1)), \tag{2.20}$$

with equality of both sides when $M(x) = 0$ understood before taking the relative error.

Proof. The prime number theorem gives $p_j \sim j \log j$ and $M(x) = o(x)$, while elementary squarefree counting gives

$$Q(x) = \frac{6}{\pi^2} x + O(\sqrt{x}) = 2c_0 x + O(\sqrt{x}). \tag{2.21}$$

Hence both $N_+(x)$ and $N_-(x)$ are $c_0 x(1 + o(1))$. If $M(x) > 0$, (2.18) is the block sum of w_j over $N_-(x) < j \le N_+(x)$; if $M(x) < 0$, it is the negative of the reversed block. Every index in that block is $c_0 x(1 + o(1))$, uniformly, and

$$w_j = p_j - j = c_0 x \log x (1 + o(1)) \tag{2.22}$$

uniformly on the block. There are $|M(x)|$ terms and their common orientation is $\operatorname{sgn} M(x)$, proving (2.20). □

3. The Dirichlet series α and its exact decomposition

Definition 3.1 (The series α and δ)**.** Define

$$\alpha(s) = \sum_{n=1}^{\infty} \frac{S(n)}{n^s}, \tag{3.1}$$

$$\delta(s) = \sum_{\substack{n \ge 1 \\ \mu(n) \neq 0}} \frac{\mu(n) w_{j(n)}}{n^s}. \tag{3.2}$$

The analytic function $\alpha(s)$ is not the ordinal position α_j introduced later.

Since $p_j \ll j\log(2j)$ and $j(n) \le n$, both series in Definition 3.1 converge absolutely in $\Re s > 2$. This is the unconditional half-plane in which termwise regrouping will first be used.

Proposition 3.2 (Four-block decomposition)**.** *For $\Re s > 2$,*

$$\alpha(s) = A(s) + B(s) + C(s) + \delta(s), \tag{3.3}$$

where

$$A(s) = \zeta(s) - \frac{\zeta(s)}{\zeta(2s)}, \tag{3.4}$$

$$B(s) = c_0 \frac{s}{(s-1)\zeta(s-1)}, \tag{3.5}$$

$$C(s) = \frac{s}{2}\int_1^\infty M(x)(Q(x) + 1 - 2c_0x)x^{-s-1}\,dx. \tag{3.6}$$

The functions A and B are meromorphic on $\mathbb{C}$. The integral defining C converges locally uniformly and defines a holomorphic function in $\Re s > 3/2$.

Proof. The nonsquarefree row of S equals

$$\sum_{\mu(n)=0} \frac{1}{n^s} = \sum_{n\ge1} \frac{1}{n^s} - \sum_{n\ge1} \frac{\mu(n)^2}{n^s} \tag{3.7}$$

$$= \zeta(s) - \frac{\zeta(s)}{\zeta(2s)} = A(s), \tag{3.8}$$

where the second equality uses the Euler product for the squarefree indicator. All series are absolutely convergent for $\Re s > 2$, in fact for this step $\Re s > 1$ suffices.

For squarefree n, the identity $p_{j(n)} = j(n) + w_{j(n)}$ gives

$$S(n) = \mu(n)j(n) + \mu(n)w_{j(n)}. \tag{3.9}$$

The second term produces δ. Let

$$J(x) = \sum_{\substack{n\le x\\ \mu(n)\ne0}} \mu(n)j(n) = \frac{M(x)(Q(x)+1)}{2}. \tag{3.10}$$

Finite Abel summation followed by passage to infinity yields

$$\sum_{\substack{n\ge1\\ \mu(n)\ne0}} \frac{\mu(n)j(n)}{n^s} = s\int_1^\infty J(x)x^{-s-1}\,dx, \tag{3.11}$$

because $J(x) = O(x^2)$ and $J(X)X^{-s} \to 0$ for $\Re s > 2$. Split $Q(x) + 1$ as $2c_0x + (Q(x) + 1 - 2c_0x)$. The remainder is exactly $C(s)$. For the main term, put $z = s - 1$ and use

$$\int_1^\infty M(x)x^{-z-1}\,dx = \frac{1}{z\zeta(z)}, \qquad \Re z > 1, \tag{3.12}$$

which follows by Abel summation from (1.6). The main term is therefore $B(s)$.

Finally, (2.21) and the unconditional bound $|M(x)| \le x$ give

$$\left|M(x)(Q(x) + 1 - 2c_0x)x^{-s-1}\right| \ll x^{1/2-\Re s}. \tag{3.13}$$

The majorant is integrable precisely when $\Re s > 3/2$. On every compact subset of that half-plane it is uniform, so dominated convergence for holomorphic parameter integrals proves holomorphy of C there. □

3.1. **Singularities and the noncancellation warning.**

Proposition 3.3 (Singularities of the explicit shifted block)**.** *If ρ is a nontrivial zero of ζ of multiplicity m, then B has a pole of order m at $s = 1+\rho$. For a simple zero,*

$$\operatorname*{Res}_{s=1+\rho} B(s) = c_0 \frac{1+\rho}{\rho\zeta'(\rho)}. \tag{3.14}$$

The function B also has a simple pole at $s = 1$ with residue $-2c_0$, poles at $s = 1-2k$ arising from the trivial zeros, and a zero rather than a pole at $s = 2$.

Proof. At a zero ρ, factor $\zeta(s-1)$ locally and note that $c_0 s/(s-1)$ is finite and nonzero at $1+\rho$. Formula (3.14) follows from the linear factor at a simple zero. At $s = 1$, use $\zeta(0) = -1/2$. At $s = 2$, the pole of $\zeta(s-1)$ makes its reciprocal vanish. The trivial zeros of $\zeta(s-1)$ occur at $s = 1-2k$. □

Warning 3.4 (No automatic poles for the full α). Proposition 3.3 locates poles of B, not of the full α. Identity (3.3) is initially established only in $\Re s > 2$. A pole of B cannot be assigned to α outside that common domain without first continuing the other terms and proving noncancellation. In particular, neither δ nor C may be silently absorbed into an unspecified holomorphic remainder.

4. Squarefree time, ordinal pairs, and finite boundaries

Let $s_1 < s_2 < \cdots$ enumerate the squarefree positive integers, and define

$$\varepsilon_q = \mu(s_q), \qquad R(q) = \sum_{r=1}^{q} \varepsilon_r = M(s_q), \qquad R(0) = 0. \tag{4.1}$$

The occurrence counters at squarefree time are

$$u_q = \frac{q + R(q)}{2}, qquad v_q = \frac{q - R(q)}{2}. \tag{4.2}$$

Let α_j and β_j denote the squarefree-time positions of the jth positive and negative signs, respectively, and put

$$a_j = s_{\alpha_j}, \qquad b_j = s_{\beta_j}, \qquad d_j = \beta_j - \alpha_j. \tag{4.3}$$

Proposition 4.1 (Pair representation)**.** *For $\Re s > 2$,*

$$\delta(s) = \sum_{j=1}^{\infty} w_j \left(a_j^{-s} - b_j^{-s} \right). \tag{4.4}$$

Moreover, for every complex s and every j,

$$a_j^{-s} - b_j^{-s} = s \int_{a_j}^{b_j} x^{-s-1}\, dx, \tag{4.5}$$

where the orientation of the integral is retained.

Proof. Absolute convergence in $\Re s > 2$ permits matching the jth positive and negative occurrences. Their residual coefficients are w_j and $-w_j$, which gives (4.4). The derivative of x^{-s} is $-sx^{-s-1}$, proving (4.5); the formula also holds at $s = 0$ by direct evaluation. □

Define the coefficient at squarefree time and its cumulative sum by

$$c_q = \varepsilon_q w_{j(s_q)}, \tag{4.6}$$

$$U(q) = \sum_{r=1}^{q} c_r = D_w(u_q) - D_w(v_q). \tag{4.7}$$

Proposition 4.2 (Open-pair block and SS readout)**.** *For every $q \geq 1$,*

$$U(q) = \operatorname{sgn} R(q) \sum_{j=\min(u_q,v_q)+1}^{\max(u_q,v_q)} w_j, \tag{4.8}$$

$$U(q) = SS(s_q) - s_q + q - \frac{q+1}{2} R(q). \tag{4.9}$$

An empty sum in (4.8) *is zero.*

Proof. The difference $D_w(u_q) - D_w(v_q)$ is the oriented block in (4.8). It consists exactly of ordinal pairs with one endpoint at or before cut q and the other endpoint after it. Formula (4.9) is (2.19) evaluated at $x = s_q$, using $Q(s_q) = q$ and $M(s_q) = R(q)$. □

Lemma 4.3 (Finite Abel identity with the endpoint term)**.** *For $Q \geq 1$ and every $s \in \mathbb{C}$, let*

$$\delta_Q(s) = \sum_{q=1}^{Q} c_q s_q^{-s}. \tag{4.10}$$

Then

$$\delta_Q(s) = U(Q) s_Q^{-s} + \sum_{q=1}^{Q-1} U(q) \left(s_q^{-s} - s_{q+1}^{-s} \right). \tag{4.11}$$

Proof. Since $c_q = U(q) - U(q-1)$, with $U(0) = 0$, finite summation gives

$$\sum_{q=1}^{Q} c_q s_q^{-s} = \sum_{q=1}^{Q} \big(U(q) - U(q-1)\big) s_q^{-s} \tag{4.12}$$

$$= U(Q) s_Q^{-s} + \sum_{q=1}^{Q-1} U(q) \big(s_q^{-s} - s_{q+1}^{-s}\big). \tag{4.13}$$

No infinite interchange occurs in this proof. □

Remark 4.4 (Meaning of the boundary). The term $U(Q) s_Q^{-s}$ is the contribution of all ordinal pairs still open at the final cut. Removing it replaces the original order by a completed-pair grouping. Such a replacement is not legitimate in the RH transfer unless the boundary is independently shown to vanish.

For $\Re s > 2$, the bound $|U(q)| \ll s_q^2 \log(2 s_q)$ makes the infinite cut representation absolutely convergent, and Lemma 4.3 then yields

$$\delta(s) = \sum_{q=1}^{\infty} U(q) \left(s_q^{-s} - s_{q+1}^{-s} \right). \tag{4.14}$$

This passage is not used to define the finite CMZ observable; the finite identity (4.11) is the primary formula.

5. The CMZ multiscale mechanism: a proof-architecture summary

5.1. **Euler vorticity and hyperbolic stretching.** The forced incompressible Euler equations and their vorticity formulation are

$$\partial_t u + (u \cdot \nabla) u + \nabla p = f, \qquad \nabla \cdot u = 0, \tag{5.1}$$

$$\partial_t \omega + (u \cdot \nabla) \omega = (\omega \cdot \nabla) u + \nabla \times f, \qquad \omega = \nabla \times u. \tag{5.2}$$

The CMZ constructions arrange an older vorticity field whose velocity is approximately linear near the origin. A younger, much smaller oscillatory packet is transported and stretched by this hyperbolic linearisation [5, 6].

In the forced non-axisymmetric construction, a parent scale N and child scale L are related by

$$L^{1/2} = exp\left(\int_{1-N^{-\varepsilon/2}}^{1} \partial_1 u_{N,1}(0,t)\,dt\right), \tag{5.3}$$

under a positive strain bound of size N^{ε}. Writing

$$K_N(t) = \exp\left(\int_t^1 \partial_1 u_{N,1}(0,s)\,ds\right), \qquad b(t) = \frac{\log K_N(t)}{\log L}, \tag{5.4}$$

the transported child packet has anisotropic widths of orders

$$L^{-1/2-b+\varepsilon}, \qquad L^{-1/2-\varepsilon+b}, \qquad L^{-1/2}, \tag{5.5}$$

and oscillation frequency of order L.

5.2. **Parametrix, cancellation, and gluing.** The proof architecture in [5] has the following linked stages.

C1. A local Biot–Savart parametrix approximates the two leading velocity components generated by the packet. For spatial derivatives of order $r = 0, 1$, the quoted error has the form

$$\|\widetilde{u}_j - u_j\|_{C^r} \le C_\kappa L^{2\varepsilon+r-3/2+\kappa}. \tag{5.6}$$

C2. The leading pieces of $\widetilde{u}_2\partial_2\omega_1$ and $\widetilde{u}_3\partial_3\omega_1$ cancel. The proof printed after the stated lemma safely yields

$$\|u(\omega)\cdot\nabla\omega\|_{C^\vartheta} + \|\omega\cdot\nabla u(\omega)\|_{C^\vartheta} \lesssim L^{4\varepsilon-1/2+\vartheta}. \tag{5.7}$$

The exponent 4ε in (5.7), rather than the 3ε appearing in the lemma statement, is the exponent obtained after the paper chooses its auxiliary loss parameter equal to ε.

C3. Oscillation and spatial separation make the child's velocity decay faster than any prescribed power outside its anisotropic support. Odd parity and Taylor expansion control interactions with the parent field.

C4. Replacing the parent velocity by its local linear model produces a controlled forcing error. With $\vartheta = 1/2 - 6\varepsilon$, the safe power in (5.7) becomes $L^{-2\varepsilon}$.

C5. The induction alternates the expanding direction and adds a child whose strain at the origin is comparable to L^{ε}, while its forcing increment is bounded by

$$\|F_{\mathrm{child}}\|_{C^{1/2-6\varepsilon}} \lesssim L^{-2\varepsilon}. \tag{5.8}$$

Choosing scales sufficiently separated makes the forcing increments summable.

These steps summarise the actual proof mechanism; they do not assert that an arbitrary prescribed time-dependent strain can be realised with constants independent of its magnitude.

5.3. **The separated-layer amplitude system.** In the unforced separated-layer construction of [6], the height $x_k(t) > 0$ of the kth packet is governed, up to controlled geometric errors, by

$$\dot{x}_k(t) = x_k(t)\sum_{j<k} a_j(t)x_j(t), \tag{5.9}$$

where $a_j(t)$ is close to 1 and, importantly, does not depend on the child index k. In the exact model $a_j = 1$, writing $y_k = \log x_k$ gives

$$\dot{y}_k(t) = \sum_{j<k} x_j(t). \tag{5.10}$$

For terminal data $x_k(0) = A^k$, $A > 1$, the model admits a uniform bound on the time integral of each amplitude over a fixed backward interval:

$$\int_{-a}^{0} x_k(t)\,dt \le a, \qquad a = A(1 - e^{-a}). \tag{5.11}$$

This positivity is structural. Encoding μ by allowing signed amplitudes would destroy the hypothesis behind (5.11). The two-channel construction in Section 6 avoids that problem by keeping both channels positive.

5.4. **Later multiscale developments.** The 2026 IPM manuscript [7] transports oscillatory packets through nested Fourier cutoffs and proves joint space–time smoothness of the force on the torus. The 2026 Boussinesq manuscript [8] uses an exact growing wave, compactly supported corrections, lifetime derivative estimates, and summable local corrections to obtain smooth forcing in both equations. These works confirm the flexibility of the multiscale programme. Their theorems concern different PDEs and do not imply an arithmetic cancellation estimate for $U(q)$.

6. The positive two-channel CMZ deformation

The construction in this section is an [ADAPT] of the logarithmic stretching law (5.10). It is an exact finite-dimensional dynamical encoding, not yet an Euler solution.

Fix a real $\sigma > 0$ and a cutoff $Q \ge 1$. Define positive clock lengths

$$\ell_q^{(Q)}(\sigma) = \begin{cases} s_q^{-\sigma} - s_{q+1}^{-\sigma}, & 1 \le q < Q, \\ s_Q^{-\sigma}, & q = Q. \end{cases} \tag{6.1}$$

Since $s_1 = 1$, these lengths satisfy

$$\sum_{q=1}^{Q} \ell_q^{(Q)}(\sigma) = 1. \tag{6.2}$$

Let $0 = \tau_0 < \tau_1 < \cdots < \tau_Q = 1$ be defined by

$$\tau_q = \sum_{r=1}^{q} \ell_r^{(Q)}(\sigma). \tag{6.3}$$

Write $U_+(q) = \max\{U(q), 0\}$ and $U_-(q) = \max\{-U(q), 0\}$. Define the nonnegative, piecewise constant strains

$$g_{\pm,Q,\sigma}(t) = U_\pm(q), \qquad \tau_{q-1} \le t < \tau_q. \tag{6.4}$$

Definition 6.1 (Paired CMZ-type channels)**.** Let $X_{\pm,Q,\sigma}$ solve

$$\dot X_{\pm,Q,\sigma}(t) = g_{\pm,Q,\sigma}(t) X_{\pm,Q,\sigma}(t), \qquad X_{\pm,Q,\sigma}(0) = 1. \tag{6.5}$$

Their relative terminal deformation is

$$\mathscr{C}_Q(\sigma) = \frac{X_{+,Q,\sigma}(1)}{X_{-,Q,\sigma}(1)}. \tag{6.6}$$

Both amplitudes in (6.5) remain positive. The arithmetic sign occurs only through their ratio.

Proposition 6.2 (Exact arithmetic readout)**.** *For every $Q \ge 1$ and every real $\sigma > 0$,*

$$\log \mathscr{C}_Q(\sigma) = \delta_Q(\sigma). \tag{6.7}$$

Proof. Solving the scalar equations gives

$$\log X_{\pm,Q,\sigma}(1) = \sum_{q=1}^{Q} U_\pm(q) \ell_q^{(Q)}(\sigma). \tag{6.8}$$

Subtracting the two identities in (6.8) yields

$$\log \mathscr{C}_Q(\sigma) = \sum_{q=1}^{Q} U(q)\ell_q^{(Q)}(\sigma). \tag{6.9}$$

Insert (6.1). The result is precisely the finite Abel identity (4.11). □

Remark 6.3 (Why real σ is used). For nonreal s, the differences $s_q^{-s} - s_{q+1}^{-s}$ are complex and cannot serve as positive time lengths. Proposition 6.2 therefore uses real σ. This loses no information in the main equivalence, because convergence of $\delta(\sigma)$ for every real $\sigma > 3/2$ is already equivalent to RH. Complex continuation is treated separately in Section 8.

7. The CMZ deformation equivalence with RH

We now state the exact bridge. It is a theorem about the explicitly defined deformation in Definition 6.1; it does not assume a PDE realisation.

Theorem 7.1 (Ordinal-prime CMZ deformation criterion)**.** *The following statements are equivalent.*

(i) *The Riemann hypothesis holds.*
(ii) *For every real $\sigma > 3/2$, the sequence $\delta_Q(\sigma)$ converges as $Q \to \infty$.*
(iii) *For every real $\sigma > 3/2$, the positive relative deformation $\mathscr{C}_Q(\sigma)$ converges to a finite nonzero limit.*
(iv) *For every real $\sigma > 3/2$,*

$$\lim_{X\to\infty} \sup_{Y\geq X} \left|\log \frac{\mathscr{C}_Y(\sigma)}{\mathscr{C}_X(\sigma)}\right| = 0. \tag{7.1}$$

Proof. Because the coefficients $c_n = \mathbf{1}_{\mu(n)\neq 0}\mu(n)w_{j(n)}$ vanish at every nonsquarefree integer, the squarefree cutoff $\delta_Q(s) = \sum_{q\leq Q} c_{s_q} s_q^{-s}$ is exactly the ordinary partial sum through the integer s_Q. Thus convergence on the squarefree clock is equivalent to convergence in the original integer ordering; no rearrangement is being made.

Assume first RH. By the Littlewood criterion (1.10), for every $\eta > 0$,

$$M(x) = O_\eta\left(x^{1/2+\eta}\right). \tag{7.2}$$

The block formula (2.18), the prime number theorem, and (7.2) imply

$$W(x) = O_\eta\left(x^{3/2+\eta}\log(2x)\right). \tag{7.3}$$

Fix $\sigma > 3/2$ and choose η with $0 < \eta < \sigma - 3/2$. Abel summation between integer cutoffs $X < Y$ gives

$$\sum_{X<n\leq Y} \frac{\mathbf{1}_{\mu(n)\neq 0}\mu(n)w_{j(n)}}{n^\sigma} = W(Y)Y^{-\sigma} - W(X)X^{-\sigma} \tag{7.4}$$

$$+ \sigma\int_X^Y W(x)x^{-\sigma-1}\,dx. \tag{7.5}$$

By (7.3), the endpoint terms tend to zero and the integral tail is dominated by a constant times

$$\int_X^\infty x^{1/2+\eta-\sigma}\log(2x)\,dx, \tag{7.6}$$

which tends to zero because $1/2 + \eta - \sigma < -1$. Thus the original-order series $\delta(\sigma)$ converges, proving (i)⇒(ii).

Conversely, suppose (ii). Fix any $\epsilon > 0$ and take

$$\sigma = \frac{3}{2} + \frac{\epsilon}{2}. \tag{7.7}$$

Convergence of $\sum c_n n^{-\sigma}$, where $c_n = \mathbf{1}_{\mu(n)\neq 0}\mu(n)w_{j(n)}$, permits Kronecker's lemma and gives

$$W(x) = o\left(x^{\sigma}\right). \tag{7.8}$$

Combine (7.8) with the two-sided asymptotic (2.20). At points where $M(x) \neq 0$ this gives

$$M(x) = o\left(\frac{x^{\sigma-1}}{\log x}\right) = o\left(\frac{x^{1/2+\epsilon/2}}{\log x}\right). \tag{7.9}$$

At points where $M(x) = 0$, the desired bound is automatic. Therefore $M(x) = O_\epsilon(x^{1/2+\epsilon})$. Since $\epsilon > 0$ was arbitrary, (1.10) gives RH. This proves (ii)⇒(i).

Proposition 6.2 gives $\mathscr{C}_Q(\sigma) = \exp(\delta_Q(\sigma))$. Since the real exponential is a homeomorphism from $\mathbb{R}$ to $(0, \infty)$, (ii) is equivalent to (iii). Finally,

$$\log \frac{\mathscr{C}_Y(\sigma)}{\mathscr{C}_X(\sigma)} = \delta_Y(\sigma) - \delta_X(\sigma), \tag{7.10}$$

so the Cauchy criterion proves the equivalence of (ii) and (iv). □

Remark 7.2 (The first RH-equivalent node). Definitions of S, SS, U, and the finite deformation are unconditional. The first statement in the dependency chain that is equivalent to RH is the tail condition (7.1). Consequently, an Euler estimate that implies this condition with an unattenuated arithmetic normalisation is an RH-strength estimate; it cannot be counted as a routine consequence of the existence of a CMZ gluing.

8. The analytic α criterion and singularity audit

Let

$$\mathcal{H} = \{s \in \mathbb{C} : \Re s > 3/2\}. \tag{8.1}$$

Theorem 8.1 (Shifted reciprocal-zeta criterion)**.** *The following are equivalent.*

(i) *RH holds.*
(ii) *B is holomorphic on $\mathcal{H}$.*
(iii) *The function $\alpha - \delta$, initially defined on $\Re s > 2$, admits a holomorphic continuation to $\mathcal{H}$.*

Proof. By (3.5), poles of B in $\mathcal{H}$ are exactly the points $1+\rho$ arising from zeros ρ of ζ with $\Re\rho > 1/2$; the elementary poles listed in Proposition 3.3 lie outside $\mathcal{H}$. Thus (i) is equivalent to (ii), using the functional-equation symmetry of nontrivial zeros.

In $\mathcal{H}$, the function C is holomorphic by Proposition 3.2. Also A is holomorphic there: its possible denominator singularities arise from zeros of $\zeta(2s)$, which have $\Re s < 1/2$, while the pole at $s = 1$ is outside $\mathcal{H}$. On $\Re s > 2$, identity (3.3) gives

$$\alpha(s) - \delta(s) = A(s) + B(s) + C(s). \tag{8.2}$$

Therefore the right-hand side extends holomorphically to $\mathcal{H}$ exactly when B does. Uniqueness follows from the identity theorem, proving the equivalence of (ii) and (iii). □

Warning 8.2 (Two distinct routes). Theorem 7.1 concerns original-order convergence of $\delta_Q(\sigma)$ on the real axis. Theorem 8.1 concerns continuation of $\alpha - \delta$. A regrouped continuation of δ does not by itself prove original-order convergence, and holomorphy of δ alone does not cancel a pole of B in the full α.

Corollary 8.3 (Abscissa statement)**.** *Let*

$$\Theta = \sup_{\zeta(\rho)=0,\ 0<\Re\rho<1} \Re\rho. \tag{8.3}$$

For the original ordering of the Dirichlet series,

$$\sigma_c(\delta) = \sigma_c(\alpha) = 1 + \Theta. \tag{8.4}$$

Proof. The classical growth-abscissa relation for M gives growth exponent Θ [3]. Proposition 2.6 raises that exponent by one for W, so the abscissa of δ is $1+\Theta$. The signed-prime part of SS has the same exponent, whereas the zero row has partial sums $O(x)$; hence α has the same abscissa. This argument is about ordinary ordering and does not assert a natural boundary or a line of poles. □

9. Conditional realisation by genuine CMZ towers

The exact criterion above uses the CMZ logarithmic law, but its prescribed piecewise strain $g_{\pm,Q,\sigma}$ is not automatically generated by a solution of the Euler equation. We isolate the additional statement needed for a genuine PDE realisation.

Definition 9.1 (Admissible CMZ readout)**.** For fixed real $\sigma > 3/2$, an admissible CMZ readout is a family of positive PDE deformation ratios $\widetilde{\mathscr{C}}_Q(\sigma)$ for which

$$\log \widetilde{\mathscr{C}}_Q(\sigma) = \delta_Q(\sigma) + r_Q(\sigma), \tag{9.1}$$

where the realisation error satisfies

$$\lim_{X\to\infty} \sup_{Y\geq X} |r_Y(\sigma) - r_X(\sigma)| = 0. \tag{9.2}$$

The coefficient of δ_Q in (9.1) is exactly one and is independent of Q and of all spatial carrier frequencies.

Theorem 9.2 (Conditional Euler-CMZ transfer)**.** *Assume that an admissible CMZ readout exists for every real $\sigma > 3/2$. Then RH is equivalent to*

$$\lim_{X\to\infty} \sup_{Y\geq X} \left| \log \frac{\widetilde{\mathscr{C}}_Y(\sigma)}{\widetilde{\mathscr{C}}_X(\sigma)} \right| = 0 \quad \textit{for every real } \sigma > 3/2. \tag{9.3}$$

Proof. Subtract (9.1) at Y and X to obtain

$$\log \frac{\widetilde{\mathscr{C}}_Y(\sigma)}{\widetilde{\mathscr{C}}_X(\sigma)} = \delta_Y(\sigma) - \delta_X(\sigma) \tag{9.4}$$

$$+ r_Y(\sigma) - r_X(\sigma). \tag{9.5}$$

Condition (9.2) shows that (9.3) holds exactly when the δ tails satisfy the Cauchy condition. Apply Theorem 7.1. □

Question 9.3 (Paired-reunion problem)**.** Can one construct two genuine CMZ towers from the ordinal pairs $+p_j, -p_j$ so that their principal Biot–Savart strains cancel after both members have appeared, while the transported reunion defects satisfy (9.2) with the exact, frequency-independent normalisation in (9.1)?

Question 9.3 is [OPEN]. Fixed-duration pairs are compatible with local transport estimates, but long pairs can undergo large hyperbolic deformation before reunion. Allowing the carrier scale to absorb $|U(q)|$ does not answer the question: it changes the readout coefficient and can make any finite datum appear small.

10. Boundary, convergence, and interchange audit

Proposition 10.1 (Complete audit of the logical dependencies)**.** *The proof has the following dependency chain:*

$$(S, SS) \longrightarrow U(q) \longrightarrow \delta_Q(\sigma) \longrightarrow \mathscr{C}_Q(\sigma) \longrightarrow \textit{tail stability} \longrightarrow \mathrm{RH}. \tag{10.1}$$

The first three arrows are finite identities. The fourth is the exact scalar ODE solution. The final arrow uses the original-order convergence theorem and the Littlewood criterion. No step before tail stability assumes RH.

Proof. The first arrow is Proposition 4.2; the second is Lemma 4.3; the third is Proposition 6.2; and the last two arrows are Theorem 7.1. □

The analytic and finite operations used above have the following status.

A1. All rearrangements defining $A + B + C + \delta$ take place first in $\Re s > 2$, where coefficient bounds give absolute convergence.
A2. Formula (4.11) is finite and valid for every complex s; it requires no convergence theorem.
A3. The physical CMZ clock uses real $\sigma > 0$, ensuring positivity of all interval lengths.
A4. The unfinished-pair boundary is retained as $U(Q)s_Q^{-\sigma}$. Under RH it vanishes by (7.3); in the converse direction it is part of δ_Q and is not separately discarded.
A5. The integral defining C is holomorphic in $\Re s > 3/2$ by the explicit majorant (3.13). No continuation of δ is used in that argument.
A6. Singularities of B are not promoted to singularities of α. The analytic criterion concerns the proved continuation of $A + B + C = \alpha - \delta$.
A7. The CMZ literature estimates concern prescribed PDE layers. They do not establish Definition 9.1; that definition is an additional hypothesis in Theorem 9.2.

Remark 10.2 (A failed shortcut). It is insufficient to prove merely that each finite arithmetic profile can be inserted into a CMZ layer after choosing an arbitrarily large carrier scale. Such a choice may make the PDE residual small while the normalised logarithmic deformation still grows with Q. The uniform, unattenuated identity (9.1) and the tail estimate (9.2) are the properties that discriminate the RH problem from finite gluing flexibility.

11. Conclusion and research route

The ordinal-prime transform produces an exact open-pair strain $U(q)$ that is readable from SS. Finite Abel summation turns this strain into the residual Dirichlet polynomial δ_Q. Splitting U into positive and negative parts permits two positive logarithmic-stretching channels, and their relative deformation equals $\exp(\delta_Q)$. Theorem 7.1 is therefore an exact dynamical reformulation of RH, with the physical clock confined to the fixed interval $[0, 1]$.

The unresolved PDE route is specific: realise the paired arithmetic strains by genuine transported CMZ packets and prove a Cauchy tail estimate for their relative deformation without allowing carrier frequencies to attenuate the arithmetic coefficient. This paired-reunion mechanism is an [OPEN] research proposal. It is not supplied by the four fluid papers cited here, and no claim of novelty relative to all literature is made.

The route count is as follows: four [LIT] multiscale inputs, one [ADAPT] two-channel stretching model, two principal [DED] bridges (finite readout and RH equivalence), and one explicitly [OPEN] PDE realisation problem.

Notebook History

Version 1.0: definitions of ζ, μ, λ, and Λ; fundamental properties of S, SS, and α; audited CMZ proof summary; finite-boundary readout; full CMZ deformation equivalence; analytic singularity audit; and conditional genuine-PDE transfer.

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